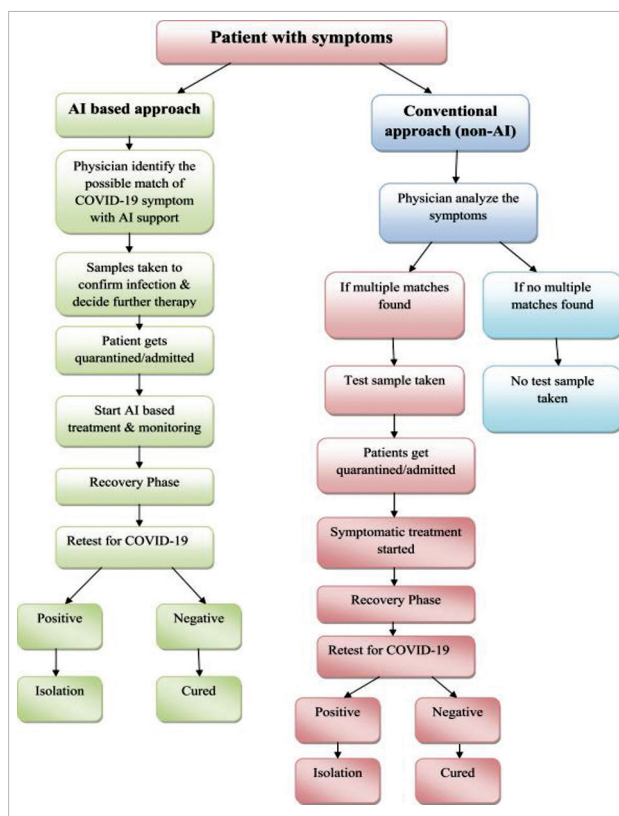


polymakers to prepare and allocate resources appropriately.

The GIS dashboard app, created by John Hopkins University, Baltimore, allows users to click on a country to identify confirmed cases and deaths. Another app, Covid-19 Surveillance Dashboard, created by the University of Virginia, provides a visualisation of Covid-19 cases, recoveries, and deaths across the globe on a real-time basis. Prediction and forecasting systems allow users to identify areas where shortages in medical equipment are likely to reach crisis point and predict surges in the cases to enable better planning for taking pre-emptive actions so that hospitals do not run out of beds, ventilators, masks, and other supplies.

Some more technologies being used to fight Covid-19

Blockchain: As Covid-19 is highly infectious, there is an urgent need for speeding up the detection of virus carriers and halting their spread. Hence, blockchain has emerged as a key technology in the critical domain of pandemic management. Blockchain technology provides a structure to store the transactional records



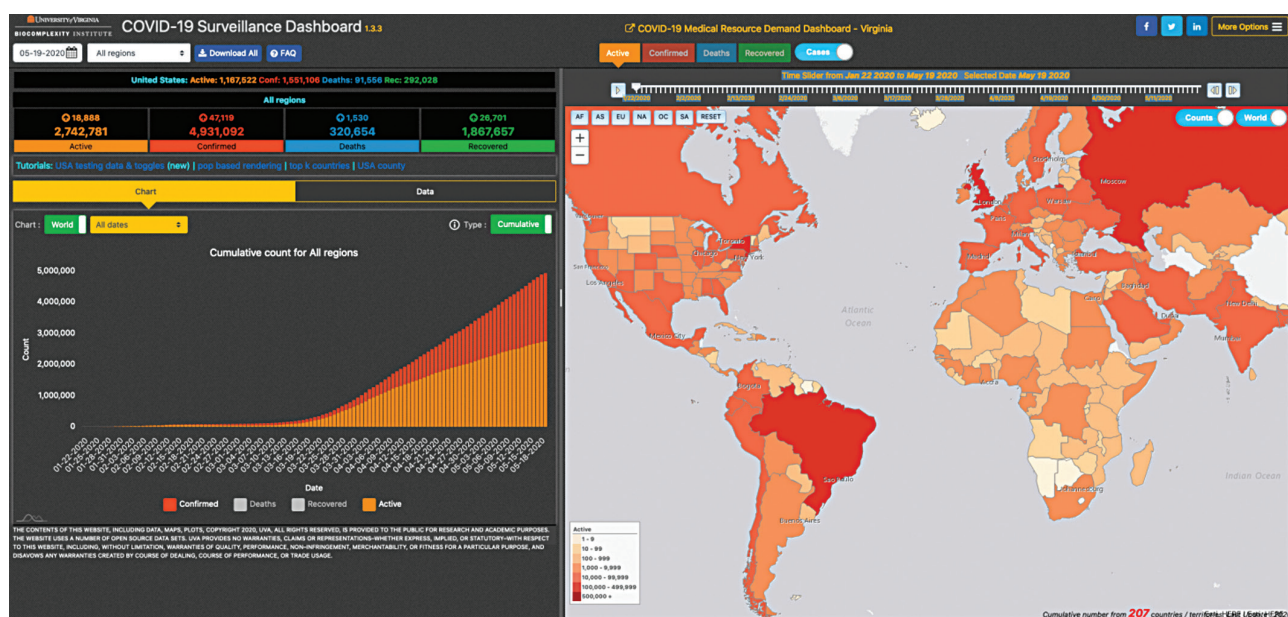
Flow chart comparing general procedure of AI and non-AI followed to identify Covid-19 symptoms along with treatments needed (Credit: www.sciencedirect.com)

in several databases in a network connected through peer-to-peer nodes, referred to as a 'digital ledger.' Every

transaction in this ledger is authenticated by the unique digital signature of the owner, safeguarding it from tampering and making the information highly secure.

Blockchain makes it possible for anyone to see the data but not tamper with it. Blockchain applications provide robust, transparent, secure, and cheap means for storing sensitive patient data and facilitating effective decision-making. Healthcare

organisations can create a centralised database and share the information with only the appropriately authorised



Covid-19 Surveillance Dashboard created by the University of Virginia (Credit: <https://covid19.biocomplexity.virginia.edu>)